

Homework 26: Eigenvalues and Eigenvectors – A Revisit

BCS A, E, F, G

Problem.

a. Reflecting on Concepts. Give short answers to each of the following statements with valid reasoning.

- i. If $\mathbf{v}$ is an eigenvector of A with corresponding eigenvalue λ and c is a scalar. Find an eigenvalue and an eigenvector of $A - cI$, A^{-1} , and A^k .
- ii. Construct a 3×3 matrix with eigenvalues 0, -1 , and 5. What conclusion can be drawn about invertibility of the constructed matrix.

b. Characteristic Polynomial or Characteristic Equation. Given a 2×2 matrix A with eigenvalues 2 and 3.

- i. Write the characteristic polynomial of A .
- ii. If A is similar to B ($A \sim B$), that is, $A = CBC^{-1}$ for some invertible matrix C . Determine the eigenvalues of B , the trace of B , and the determinant of B .

c. Eigenvalues and Eigenspaces of Linear Transformations. By inspection, find the eigenvalues and corresponding eigenspaces of the following linear transformations in $\mathbb{R}^2$.

- i. Reflection about the line $y = x$.
- ii. Projection on the x -axis.
- iii. Contraction with factor k ($0 \leq k < 1$).

Note. “By inspection”: You are not required to determine the matrix of linear transformation.

Submission Date: October 25, 2024 (Friday – 11:59PM)

Good Luck